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mysql> USE string_functions_db;
Database changed
mysql>
mysql> -- -----
mysql> -- 1. SPACE() - Returns a string of specified number of spaces
mysql> -- -----
mysql> SELECT CONCAT('A', SPACE(5), 'B') AS Space_Result;
+-----+
| Space_Result |
+-----+
| A      B      |
+-----+
1 row in set (0.00 sec)

mysql>
mysql> -- -----
mysql> -- 2. SUBSTRING() - Extracts characters from a specific position
mysql> -- -----
mysql> SELECT SUBSTRING('Database Management System', 1, 8) AS Substring_Result;
+-----+
| Substring_Result |
+-----+
| Database         |
+-----+
1 row in set (0.00 sec)

mysql>
mysql> -- -----
mysql> -- 3. SUBSTR() - Synonym for SUBSTRING()
mysql> -- -----
mysql> SELECT SUBSTR('Database Management System', 10, 10) AS Substr_Result;
+-----+
| Substr_Result |
+-----+
| Management    |
+-----+
1 row in set (0.00 sec)

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mysql> -- -----
mysql> -- 4. UPPER() - Converts string to uppercase
mysql> -- -----
mysql> SELECT UPPER('mysql string functions') AS Upper_Result;
+-----+
| Upper_Result          |
+-----+
| MYSQL STRING FUNCTIONS |
+-----+
1 row in set (0.00 sec)

mysql>
mysql> -- -----
mysql> -- 5. LOWER() - Converts string to lowercase
mysql> -- -----
mysql> SELECT LOWER('MYSQL STRING FUNCTIONS') AS Lower_Result;
+-----+
| Lower_Result          |
+-----+
| mysql string functions |
+-----+
1 row in set (0.00 sec)

mysql>
mysql> -- -----
mysql> -- 6. LCASE() - Synonym for LOWER()
mysql> -- -----
mysql> SELECT LCASE('MYSQL STRING FUNCTIONS') AS Lcase_Result;
+-----+
| Lcase_Result          |
+-----+
| mysql string functions |
+-----+
1 row in set (0.00 sec)

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mysql>
mysql> -- -----
mysql> -- 7. TRIM() - Removes leading and trailing spaces/characters
mysql> -- -----
mysql> SELECT TRIM('   MySQL Trim Example   ') AS Trim_Result;
+-----+
| Trim_Result |
+-----+
| MySQL Trim Example |
+-----+
1 row in set (0.00 sec)

mysql> SELECT TRIM(BOTH 'x' FROM 'xxxMySQLxxx') AS Trim_Both_Result;
+-----+
| Trim_Both_Result |
+-----+
| MySQL |
+-----+
1 row in set (0.00 sec)

mysql> SELECT TRIM(LEADING 'x' FROM 'xxxMySQLxxx') AS Trim_Leading_Result;
+-----+
| Trim_Leading_Result |
+-----+
| MySQLxxx |
+-----+
1 row in set (0.00 sec)

mysql> SELECT TRIM(TRAILING 'x' FROM 'xxxMySQLxxx') AS Trim_Trailing_Result;
+-----+
| Trim_Trailing_Result |
+-----+
| xxxMySQL |
+-----+
1 row in set (0.00 sec)

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1 row in set (0.00 sec)

mysql>

mysql> -- -----

mysql> -- 8. RTRIM() - Removes trailing spaces

mysql> -- -----

mysql> SELECT CONCAT(RTRIM('MySQL '), '|') AS Rtrim_Result;

+-----+

| Rtrim_Result |

+-----+

| MySQL| |

+-----+

1 row in set (0.00 sec)

mysql>

mysql> -- -----

mysql> -- 9. LTRIM() - Removes leading spaces

mysql> -- -----

mysql> SELECT CONCAT('|', LTRIM(' MySQL')) AS Ltrim_Result;

+-----+

| Ltrim_Result |

+-----+

| |MySQL |

+-----+

1 row in set (0.00 sec)

mysql>

mysql> -- -----

mysql> -- 10. LENGTH() - Returns length of string in bytes

mysql> -- -----

mysql> SELECT LENGTH('MySQL') AS Length_Result;

+-----+

| Length_Result |

+-----+

| 5 |

+-----+

1 row in set (0.00 sec)

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mysql>
mysql> -- -----
mysql> -- 11. BIT_LENGTH() - Returns length of string in bits
mysql> -- -----
mysql> SELECT BIT_LENGTH('MySQL') AS Bit_Length_Result;
+-----+
| Bit_Length_Result |
+-----+
|                40 |
+-----+
1 row in set (0.00 sec)

mysql>
mysql> -- -----
mysql> -- 12. CHAR_LENGTH() - Returns length of string in characters
mysql> -- -----
mysql> SELECT CHAR_LENGTH('MySQL') AS Char_Length_Result;
+-----+
| Char_Length_Result |
+-----+
|                   5 |
+-----+
1 row in set (0.00 sec)

mysql>
mysql> -- -----
mysql> -- 13. CONCAT() - Concatenates two or more strings
mysql> -- -----
mysql> SELECT CONCAT('My', 'SQL', ' ', 'Server') AS Concat_Result;
+-----+
| Concat_Result |
+-----+
| MySQL Server  |
+-----+
1 row in set (0.00 sec)

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mysql> -- -----
mysql> -- 14. INSTR() - Returns position of first occurrence of substring
mysql> -- -----
mysql> SELECT INSTR('Database Management', 'Management') AS Instr_Result;
+-----+
| Instr_Result |
+-----+
|          10 |
+-----+
1 row in set (0.00 sec)

mysql>
mysql> -- -----
mysql> -- 15. LOCATE() - Returns position of substring within string
mysql> -- -----
mysql> SELECT LOCATE('Management', 'Database Management') AS Locate_Result;
+-----+
| Locate_Result |
+-----+
|          10 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT LOCATE('a', 'Database Management', 5) AS Locate_With_Pos_Result;
+-----+
| Locate_With_Pos_Result |
+-----+
|              6 |
+-----+
1 row in set (0.00 sec)

mysql>
mysql> -- -----
mysql> -- 16. MID() - Extracts a substring (synonym for SUBSTRING)
mysql> -- -----
mysql> SELECT MID('Database Management System', 10, 10) AS Mid_Result;
+-----+
| Mid_Result |
+-----+
| Management |
+-----+

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mysql> -- 16. MID() - Extracts a substring (synonym for SUBSTRING)
mysql> -- -----
mysql> SELECT MID('Database Management System', 10, 10) AS Mid_Result;
+-----+
| Mid_Result |
+-----+
| Management |
+-----+
1 row in set (0.00 sec)

mysql>
mysql> -- -----
mysql> -- 17. POSITION() - Returns position of substring within string
mysql> -- -----
mysql> SELECT POSITION('Management' IN 'Database Management') AS Position_Result;
+-----+
| Position_Result |
+-----+
| 10 |
+-----+
1 row in set (0.00 sec)

mysql>
mysql> -- -----
mysql> -- ALL FUNCTIONS TOGETHER IN A SINGLE RESULT (for one-shot demo)
mysql> -- -----
mysql> SELECT
-> CONCAT('A', SPACE(5), 'B') AS Space_Result,
-> SUBSTRING('Database Management System', 1, 8) AS Substring_Result,
-> SUBSTR('Database Management System', 10, 10) AS Substr_Result,
-> UPPER('mysql string functions') AS Upper_Result,
-> LOWER('MYSQL STRING FUNCTIONS') AS Lower_Result,
-> TRIM(' MySQL Trim Example ') AS Trim_Result,
-> RTRIM('MySQL ') AS Rtrim_Result,
-> LTRIM(' MySQL') AS Ltrim_Result,
-> LENGTH('MySQL') AS Length_Result,
-> BIT_LENGTH('MySQL') AS Bit_Length_Result,
-> CHAR_LENGTH('MySQL') AS Char_Length_Result,
-> CONCAT('My', 'SQL', ' ', 'Server') AS Concat_Result,
-> INSTR('Database Management', 'Management') AS Instr_Result,
-> LOCATE('Management', 'Database Management') AS Locate_Result,
+-----+
| Position_Result |
+-----+
| 10 |
+-----+
1 row in set (0.00 sec)

mysql>
mysql> -- -----
mysql> -- ALL FUNCTIONS TOGETHER IN A SINGLE RESULT (for one-shot demo)
mysql> -- -----
mysql> SELECT
-> CONCAT('A', SPACE(5), 'B') AS Space_Result,
-> SUBSTRING('Database Management System', 1, 8) AS Substring_Result,
-> SUBSTR('Database Management System', 10, 10) AS Substr_Result,
-> UPPER('mysql string functions') AS Upper_Result,
-> LOWER('MYSQL STRING FUNCTIONS') AS Lower_Result,
-> TRIM(' MySQL Trim Example ') AS Trim_Result,
-> RTRIM('MySQL ') AS Rtrim_Result,
-> LTRIM(' MySQL') AS Ltrim_Result,
-> LENGTH('MySQL') AS Length_Result,
-> BIT_LENGTH('MySQL') AS Bit_Length_Result,
-> CHAR_LENGTH('MySQL') AS Char_Length_Result,
-> CONCAT('My', 'SQL', ' ', 'Server') AS Concat_Result,
-> INSTR('Database Management', 'Management') AS Instr_Result,
-> LOCATE('Management', 'Database Management') AS Locate_Result,
-> MID('Database Management System', 10, 10) AS Mid_Result,
-> POSITION('Management' IN 'Database Management') AS Position_Result;
+-----+
| Space_Result | Substring_Result | Substr_Result | Upper_Result | Lower_Result | Trim_Result | Rtrim_Result | Ltrim_Result | Len |
+-----+
| A | B | Database | Management | MYSQL STRING FUNCTIONS | mysql string functions | MySQL Trim Example | MySQL | MySQL |
+-----+
| 5 | 40 | 5 | 10 | 10 | Management | 10 | | |
+-----+
1 row in set (0.00 sec)

mysql> |

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